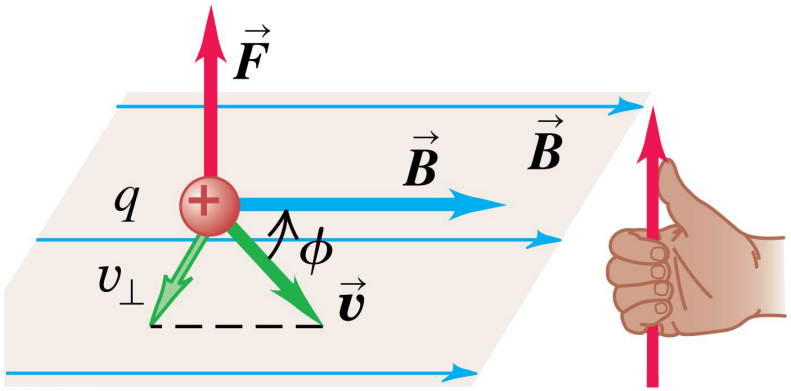
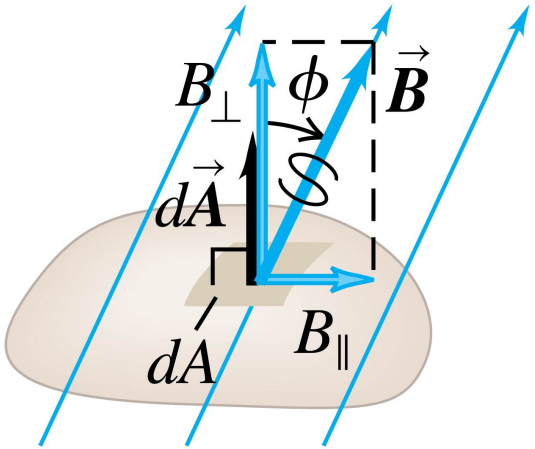
In the previous lecture

- Magnetic forces $\vec{F} = q\vec{v} \times \vec{B}$

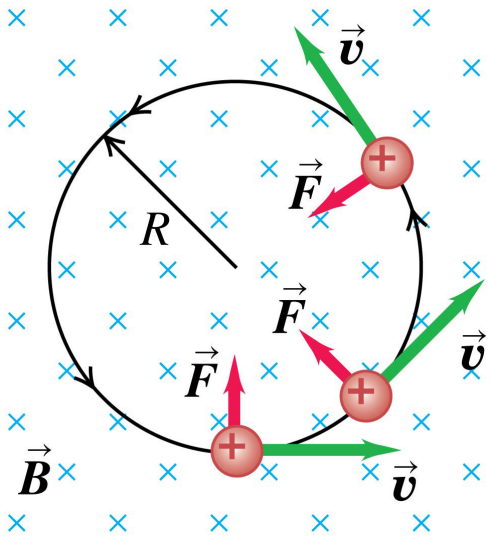


- Magnetic field lines & flux



- Motion in a magnetic field

$$R = \frac{mv}{|q|B}$$



$$\Phi_B = \int B\cos\phi\, dA$$

$$= \int B_\perp\, dA$$

$$= \int \vec{B} \cdot d\vec{A}$$

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad \text{(closed surface)}$$

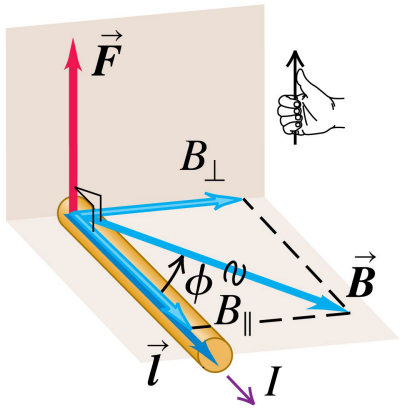
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In the previous lectures

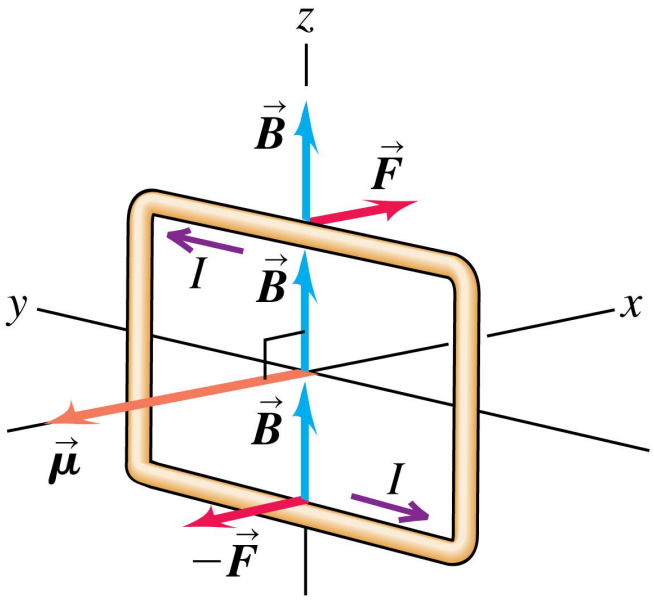
- Magnetic forces on a conductor

$$\vec{F} = I\vec{l} \times \vec{B}$$

$$d\vec{F} = I\, d\vec{l} \times \vec{B}$$



- Magnetic torque

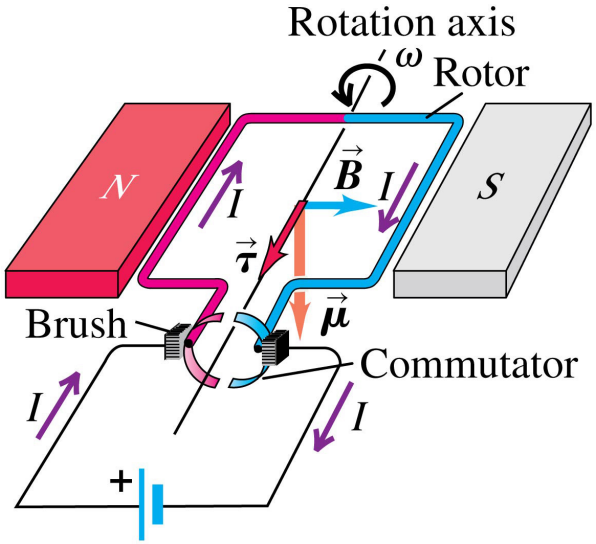


$$\tau = IBA\sin\phi$$

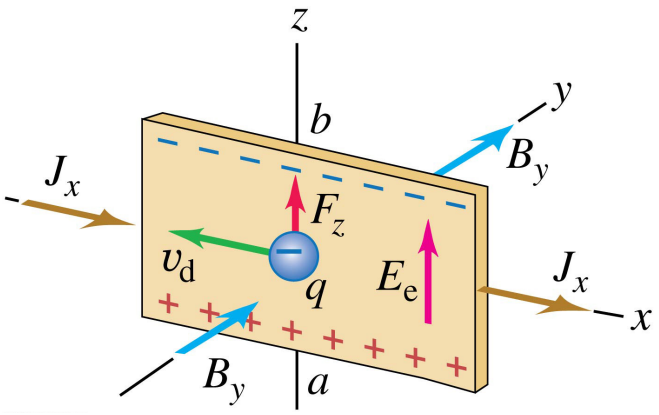
$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$U = -\vec{\mu} \cdot \vec{B} = -\mu B\cos\phi$$

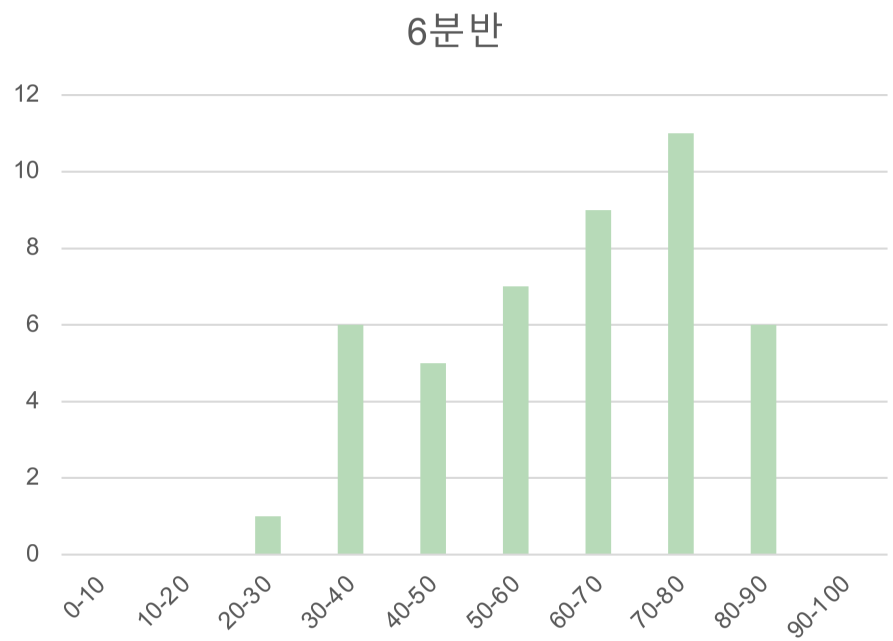
- Electric motors



- Hall effect



중간고사 결과



6분반

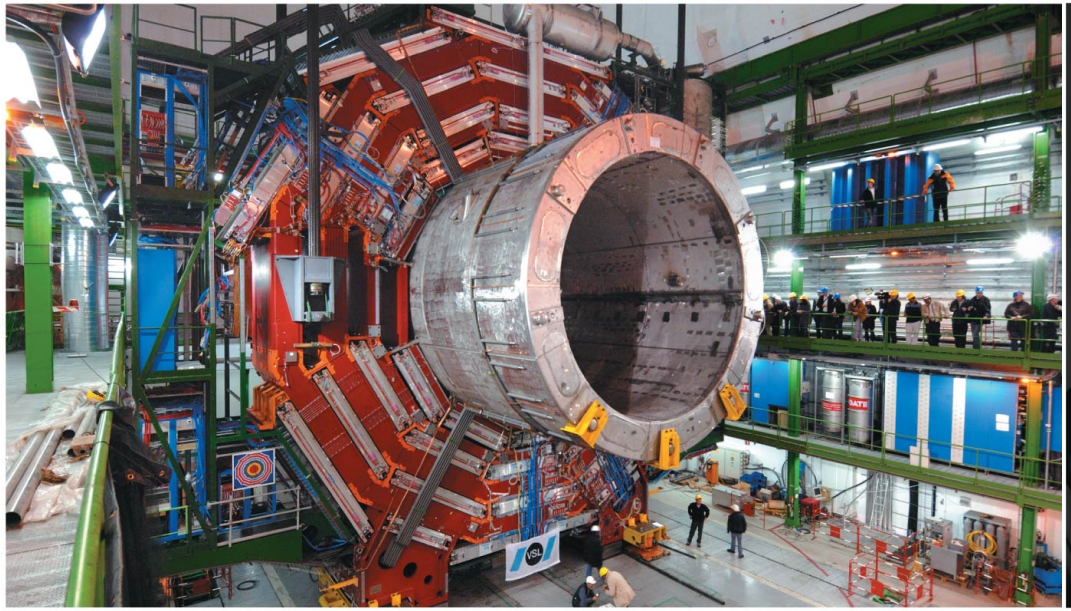
점수	명
0-10	0
10-20	0
20-30	1
30-40	6
40-50	5
50-60	7
60-70	9
70-80	11
80-90	6
90-100	0
총합	**45**
평균	**61.28**
표준편차	**16.79**

중간고사 합계 (100)

점수	명
0-10	8
10-20	7
20-30	17
30-40	29
40-50	47
50-60	61
60-70	81
70-80	47
80-90	31
90-100	0
총합	**328**
평균	**56.11**
표준편차	**18.60**

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Ch28 Sources of Magnetic Field



CERN, the European Organization for Nuclear Research.



MRI (magnetic resonance imaging)







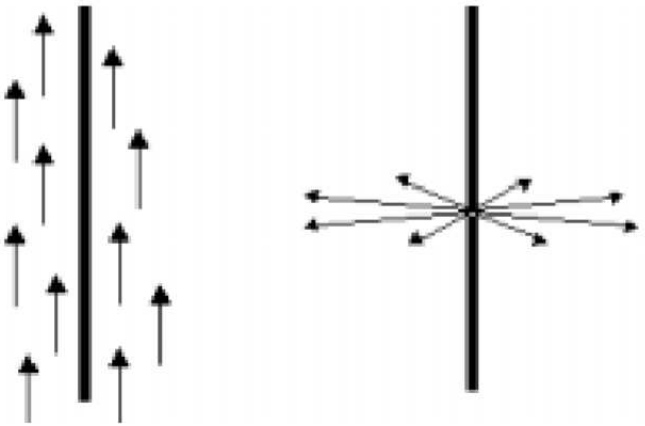
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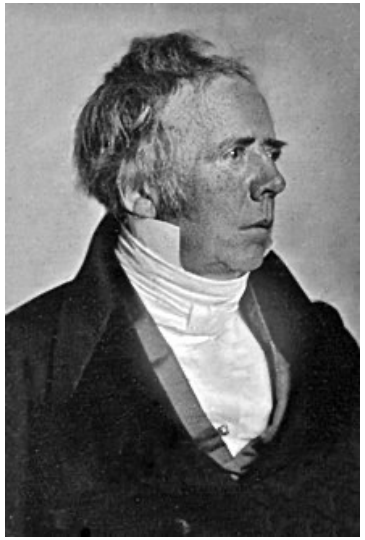
Goals for Chapter 28

- Magnetic field of a moving charged particle
- Magnetic field of a current element – **the law of Biot and Savart**
- Magnetic field of a long straight current-carrying wire
- Force between current-carrying wires
- Magnetic field of a current loop
- **Ampere's law and its application**

Question 15

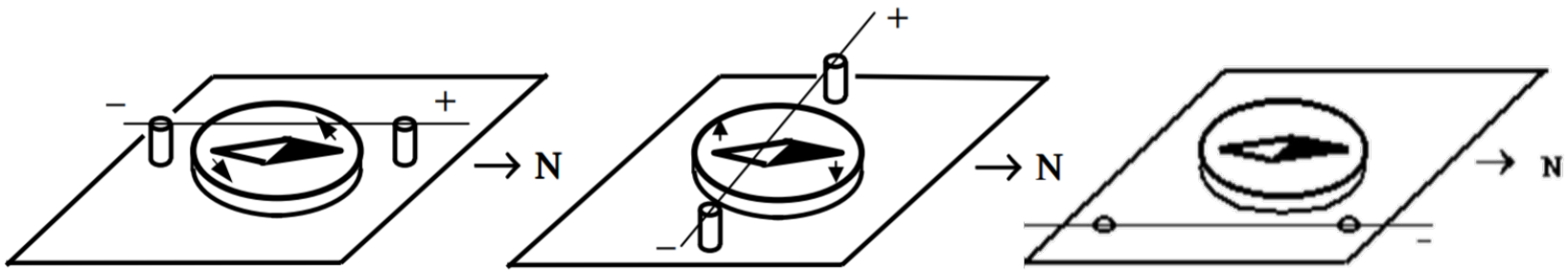
- **In Ørsted's experiments (1820) using a current-carrying wire and a compass needle, he conjectured the following hypothetical magnetic field.**





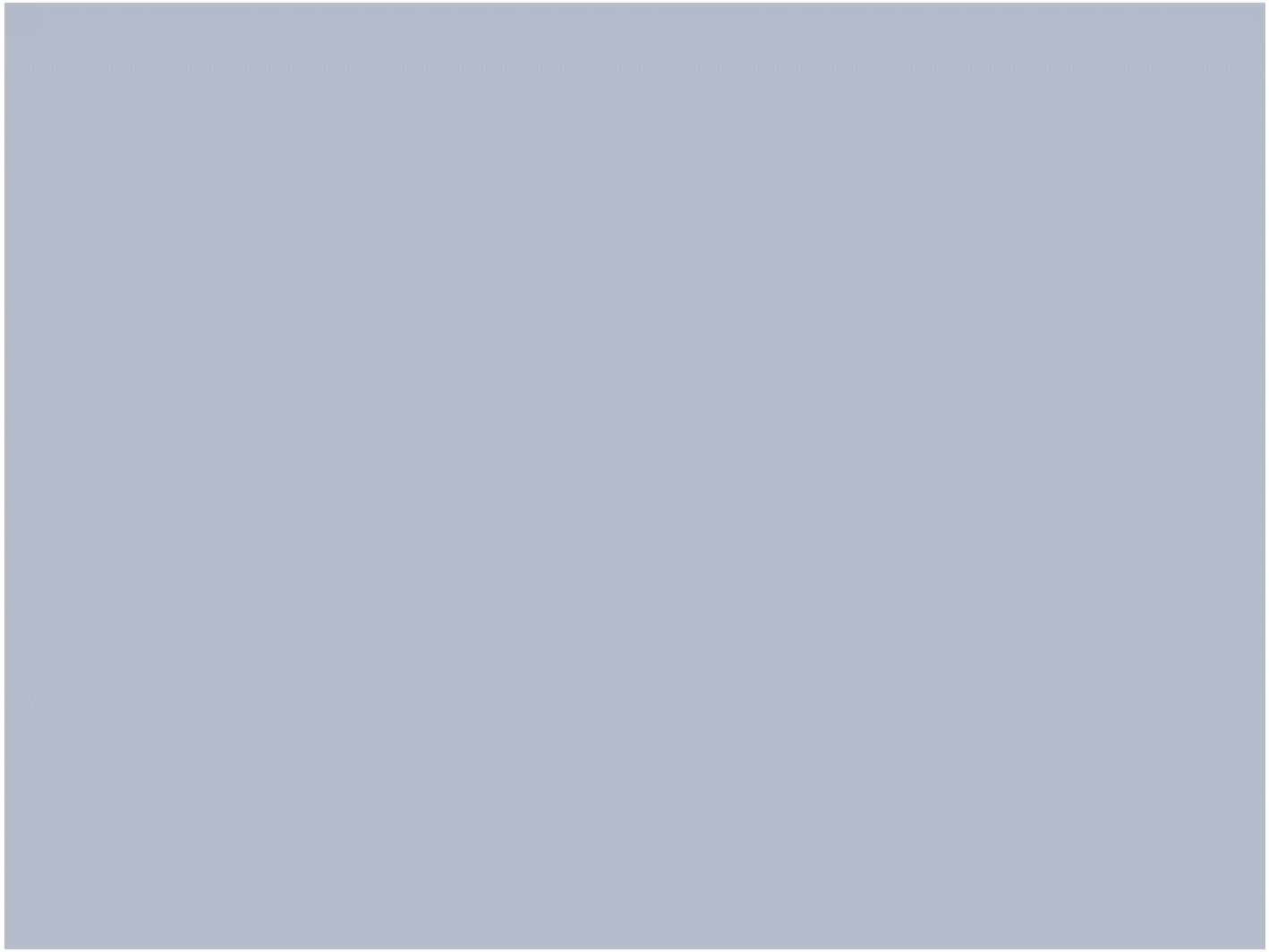
Hans Christian Ørsted

What would be the most appropriate experimental set up to confirm the hypothesis?



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The magnetic field of a moving charge

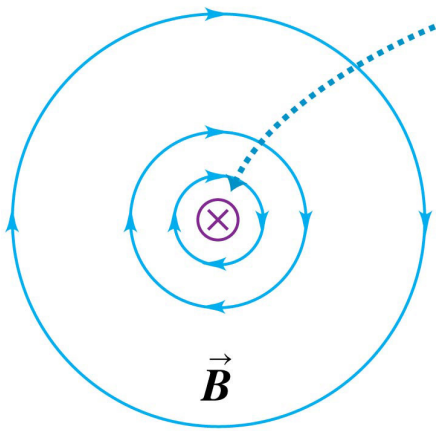


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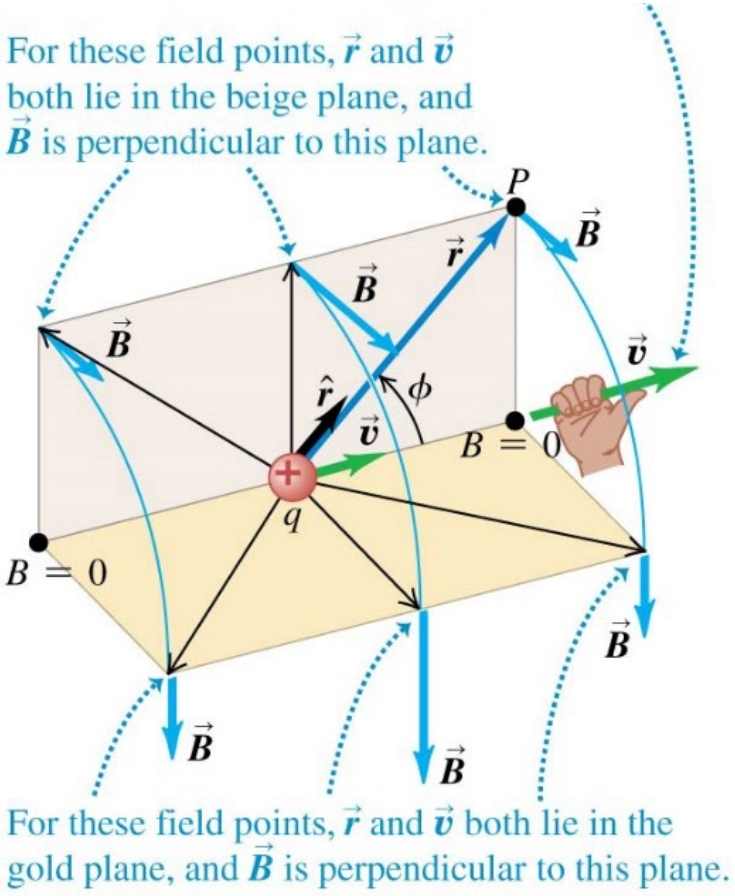
The magnetic field of a moving charge

- A moving charge generates a magnetic field that depends on the velocity of the charge, and the distance from the charge.

View from behind the charge



The $\times$ symbol indicates that the charge is moving into the plane of the page (away from you).



Magnetic field due to a point charge with constant velocity

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$$

μ_0: Magnetic constant; q: Charge; $\vec{v}$: Velocity; $\hat{r}$: Unit vector from point charge toward where field is measured; r: Distance from point charge to where field is measured

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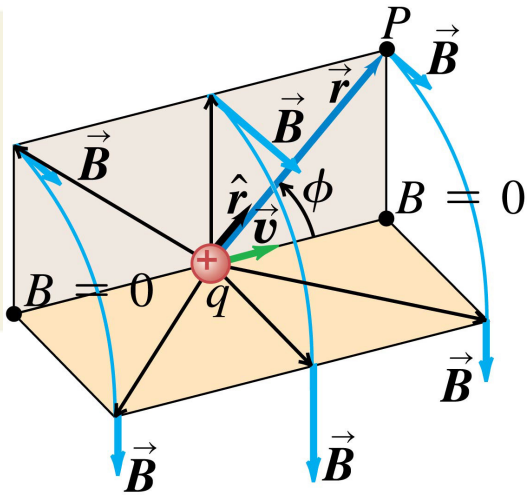
Principle of superposition of magnetic fields

- The total magnetic field caused by several moving charges is the vector sum of the fields caused by the individual charges.

Magnetic field due to a point charge with constant velocity

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$$

Magnetic constant; Charge; Velocity; Unit vector from point charge toward where field is measured; Distance from point charge to where field is measured



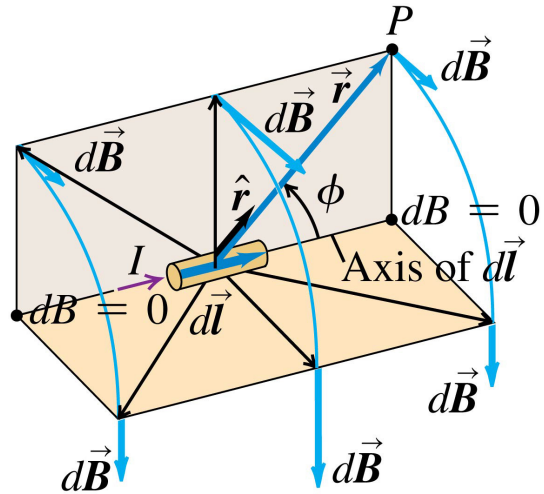
$$N = nAdl$$

$$Nq\vec{v}_d = nq\vec{v}_dAdl = \vec{J}Adl = Id\vec{l}$$

Magnetic field due to an infinitesimal current element

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l}\times\hat{r}}{r^2}$$

Magnetic constant; Current; Vector length of element (points in current direction); Unit vector from element toward where field is measured; Distance from element to where field is measured



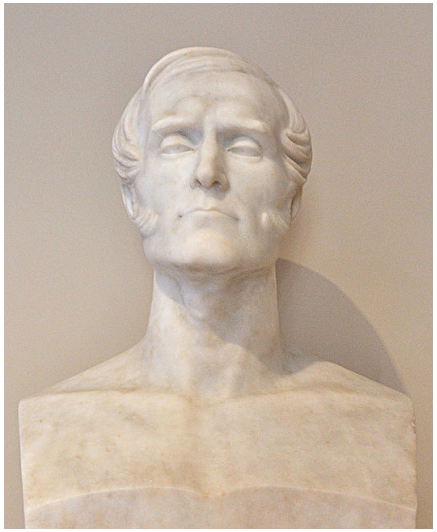
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Magnetic field of a current element

- The total magnetic field of several moving charges is the vector sum of each field.



Jean-Baptiste Biot
(1774-1862)



Félix Savart
(1791- 1841)

- The magnetic field caused by a short segment of a current-carrying conductor is found using the **law of Biot and Savart**:

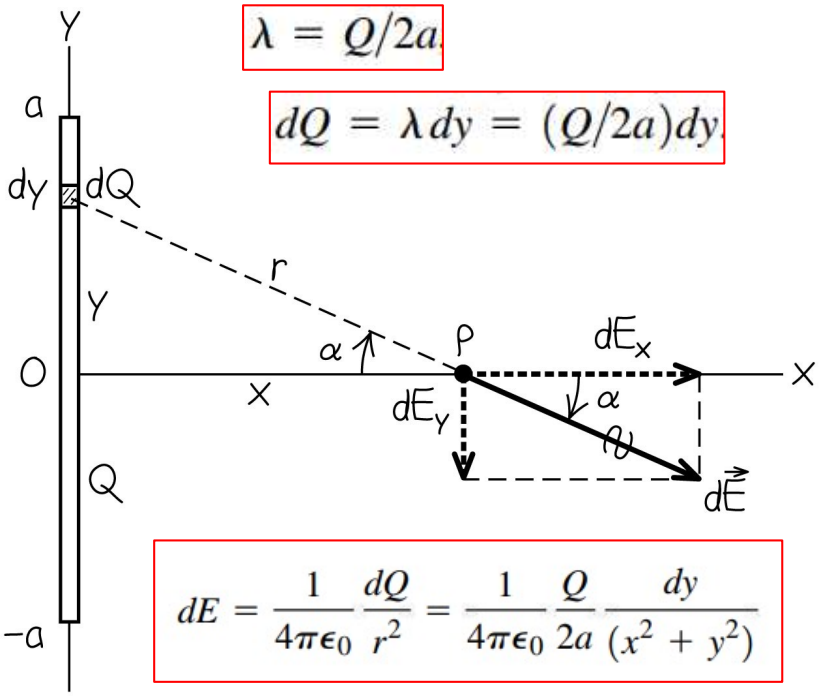
$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l}\times\hat{r}}{r^2}$$

$$\vec{B} = \frac{\mu_0}{4\pi}\int\frac{I\,d\vec{l}\times\hat{r}}{r^2}$$

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Ex 21.10: Field of a charged line segment

Positive charge Q is distributed uniformly along the y-axis between $y = -a$ and $y = +a$. Find the electric field at point P on the x-axis at a distance x from the origin.



$$\lambda = Q/2a$$

$$dQ = \lambda\, dy = (Q/2a)dy$$

$$dE_x = dE\cos\alpha = \frac{1}{4\pi\epsilon_0}\frac{Q}{2a}\frac{x\,dy}{(x^2+y^2)^{3/2}}$$

$$dE_y = -dE\sin\alpha = -\frac{1}{4\pi\epsilon_0}\frac{Q}{2a}\frac{y\,dy}{(x^2+y^2)^{3/2}}$$

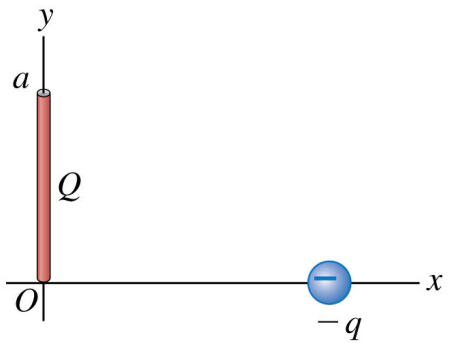
$$dE = \frac{1}{4\pi\epsilon_0}\frac{dQ}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{Q}{2a}\frac{dy}{(x^2+y^2)}$$

$$E_x = \frac{1}{4\pi\epsilon_0}\frac{Q}{2a}\int_{-a}^{+a}\frac{x\,dy}{(x^2+y^2)^{3/2}} = \frac{Q}{4\pi\epsilon_0}\frac{1}{x\sqrt{x^2+a^2}}$$

$$E_y = -\frac{1}{4\pi\epsilon_0}\frac{Q}{2a}\int_{-a}^{+a}\frac{y\,dy}{(x^2+y^2)^{3/2}} = 0$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{x\sqrt{x^2+a^2}}\hat{\imath}$$

$x \gg a$, $\vec{E} = \dfrac{1}{4\pi\epsilon_0}\dfrac{Q}{x^2}\hat{\imath}$



$a \gg x$, $\vec{E} = \dfrac{1}{2\pi\epsilon_0}\dfrac{\lambda}{x\sqrt{(x^2/a^2)+1}}\hat{\imath} = \dfrac{\lambda}{2\pi\epsilon_0 x}\hat{\imath}$

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Magnetic field of a straight current-carrying conductor

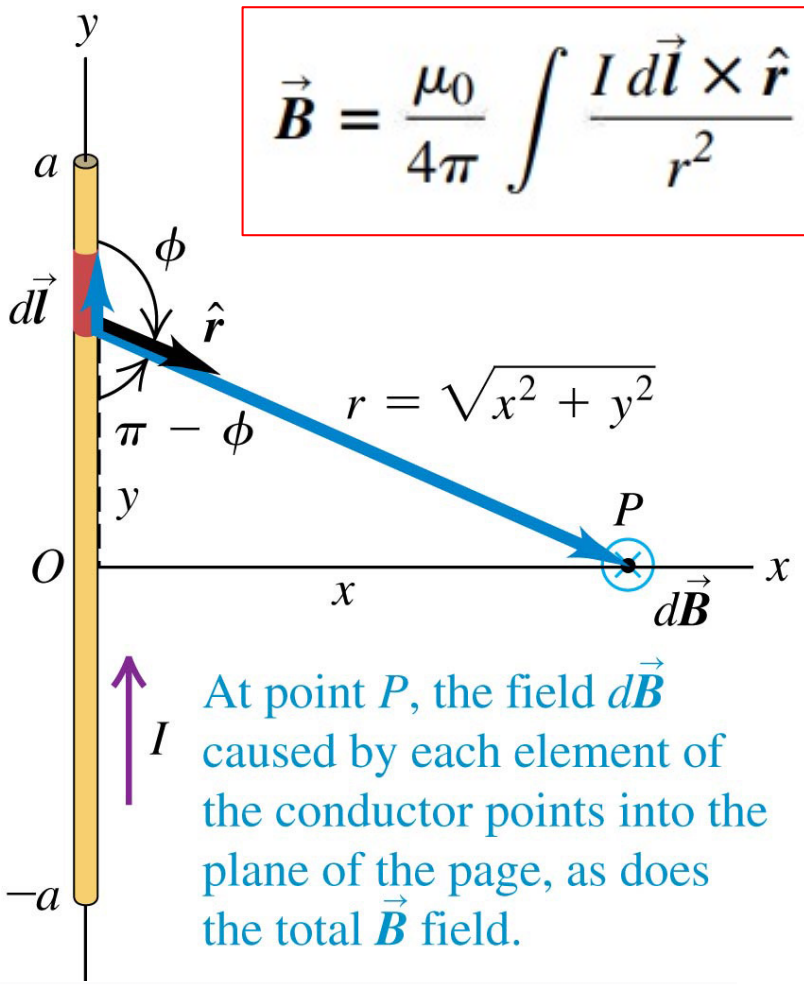
$$B = \frac{\mu_0 I}{4\pi} \int_{-a}^{a} \frac{x\,dy}{(x^2 + y^2)^{3/2}}$$

$$B = \frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2 + a^2}}$$

in the limit $a \to \infty$,

$$B = \frac{\mu_0 I}{2\pi x}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I\,d\vec{l} \times \hat{r}}{r^2}$$



At point P, the field $d\vec{B}$ caused by each element of the conductor points into the plane of the page, as does the total $\vec{B}$ field.

Magnetic field near a long, straight, current-carrying conductor

$$B = \frac{\mu_0 I}{2\pi r}$$

μ_0: Magnetic constant; I: Current; r: Distance from conductor

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Magnetic field of a straight current-carrying conductor

- The field lines around a long, straight, current-carrying conductor are circles, with directions determined by the right-hand rule.

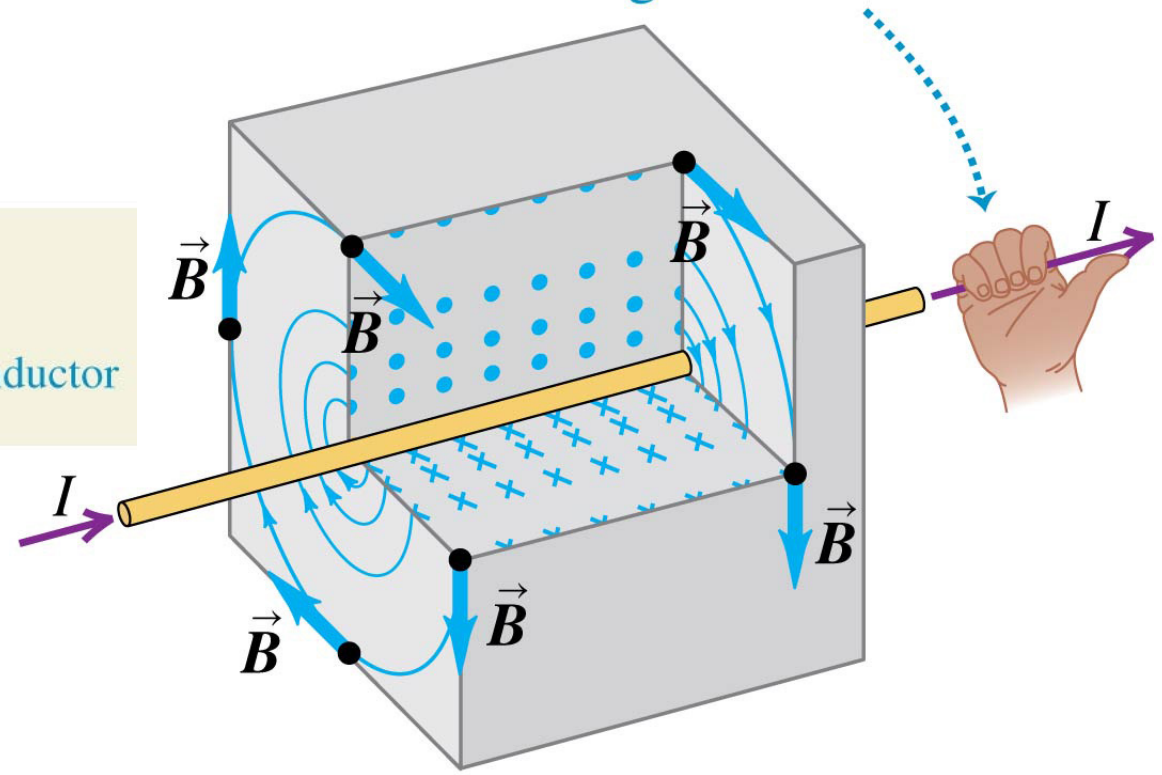
Right-hand rule for the magnetic field around a current-carrying wire: Point the thumb of your right hand in the direction of the current. Your fingers now curl around the wire in the direction of the magnetic field lines.

Magnetic constant

$$B = \frac{\mu_0 I}{2\pi r}$$

Current

Distance from conductor



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